

Multiterminal Pathfinding in Practical VLSI Systems with Deep Neural Networks

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A multiterminal obstacle-avoiding pathfinding approach is proposed. The approach is inspired by deep image learning. The key idea is based on training a conditional generative adversarial network (cGAN) to interpret a pathfinding task as a graphical bitmap and consequently map a pathfinding task onto a pathfinding solution represented by another bitmap. To enable the proposed cGAN pathfinding, a methodology for generating synthetic dataset is also proposed. The cGAN model is implemented in Python/Keras, trained on synthetically generated data, evaluated on practical VLSI benchmarks, and compared with state-of-the-art. Due to effective parallelization on GPU hardware, the proposed approach yields a state-of-the-art-like wirelength and a better runtime and throughput for moderately complex pathfinding tasks. However, the runtime and throughput with the proposed approach remain constant with an increasing task complexity, promising orders of magnitude improvement over state-of-the-art in complex pathfinding tasks. The cGAN pathfinder can be exploited in numerous high throughput applications, such as, navigation, tracking, and routing in complex VLSI systems. The last is of particular interest to this work.

CCS Concepts: • **Hardware** → **Physical design (EDA)**; • **Computing methodologies** → **Neural networks**;

Additional Key Words and Phrases: Computer-aided design, CAD, electronic design automation, EDA, pathfinding, global routing, machine learning, deep learning, deep neural networks, DeconvNets, convolutional neural networks, ConvNets, generative adversarial networks, GANs, variational autoencoders, VAEs

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1 INTRODUCTION

Multiterminal pathfinding in presence of obstacles, or **obstacle-avoiding rectilinear Steiner minimum tree construction (OARSMT)** is a task of finding an optimal (e.g., shortest) path between two or more placed terminals. While being computationally hard, this is one of the fundamental navigation problems, required in a wide range of applications, from mobile robot path planning, navigation, wayfinding, and tracking to routing wires in **printed circuit boards (PCBs)** and **integrated circuits (ICs)**. In particular, multiterminal pathfinding in presence of obstacles plays an important role in **electronic design automation (EDA)** algorithms, such as global routing and placement. Furthermore, recent trends in **machine learning (ML)**-based EDA require an

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